NEXASTORE

AI Machines • Developer Workstations • Next-Generation Computing



NPU vs GPU

NPU vs GPU for AI Workloads

Choose the right accelerator for the job

NexaStore Knowledge Series • 2026 Edition

1. Two Accelerators, Different Strengths

NPUs and GPUs can both accelerate AI, but they are optimized for different operating envelopes. An NPU emphasizes efficient, low-power on-device inference for supported workloads. A discrete GPU provides broad parallel compute and is commonly preferred for local LLMs, computer vision, generative AI experimentation and many established ML frameworks.

Dimension	NPU	Discrete GPU
Primary goal	Efficient on-device AI	High-throughput parallel compute
Power efficiency	Excellent	Lower under sustained heavy load
Battery-oriented use	Strong fit	Can reduce battery life quickly
Local LLM ecosystem	Runtime/model dependent and evolving	Broad support across many mature runtimes
Computer vision development	Supported in selected stacks	Strong general-purpose option
Training / fine-tuning	Limited use cases	Much stronger fit
Memory model	Often shared/system-integrated	Dedicated VRAM on discrete GPUs

2. When an NPU Makes Sense

• Always-on AI features where battery efficiency matters.

• Supported speech, camera, summarization or productivity workloads optimized for the NPU.

• Enterprise users who primarily consume cloud AI but want efficient local AI features.

• Thin-and-light systems where sustained discrete-GPU power draw is undesirable.

3. When a GPU Makes Sense

• Frequent local LLM inference where the chosen runtime supports GPU acceleration.

• CUDA-oriented AI/ML development and computer vision.

• Generative image, video or creator workflows.

• Fine-tuning experiments and workloads requiring substantial dedicated VRAM.

4. Workload Decision Matrix

Workload	NPU fit	GPU fit	Recommendation direction
Cloud AI API development	Optional	Optional	Prioritize CPU, RAM and battery; accelerator is secondary.
On-device productivity AI	Excellent	Good	Prefer NPU when application/runtime is optimized for it.
Local 7B-14B LLMs	Variable	Excellent	GPU usually provides the broader ecosystem and VRAM options.
Computer vision prototyping	Variable	Excellent	GPU is typically the safer development choice.
RAG with cloud embeddings	Optional	Optional	RAM/CPU/storage often matter more.
RAG with local embeddings	Good if supported	Excellent	Choose based on embedding runtime support and batch needs.
Model fine-tuning	Limited	Excellent	High-VRAM GPU or cloud accelerator is preferred.

5. Do Not Compare TOPS Alone

TOPS is a useful hardware metric only within context. Different precisions, architectures, software stacks and supported operators make headline TOPS numbers poor substitutes for application-level testing. A recommendation agent should not rank an NPU and GPU solely by advertised TOPS.

6. NexaStore Recommendation Rule

Use the product catalog for exact accelerator, VRAM, NPU and SKU facts. Use this guide for suitability reasoning. If a buyer names a specific framework or runtime, compatibility should be checked before recommending hardware.

Educational notice

This document is part of the fictional NexaStore educational case study.